Tennis Analysis

This notebook evaluates the three research goals:

1. How important is ranking for match outcomes?
2. Do players perform differently depending on the surface?
3. Which player characteristics are associated with winning matches?

The analysis uses:

- `data/matches_clean.csv`
- `data/stats_clean.csv`
- `data/top20_players_raw.csv`

In [52]:
```
import pandas as pd
import numpy as np
import os
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import matplotlib.colors as mcolors

PLOTS_DIR = "plots"
os.makedirs(PLOTS_DIR, exist_ok=True)

# import of CSV files
matches = pd.read_csv("data/matches_clean.csv")
stats = pd.read_csv("data/stats_clean.csv")
players = pd.read_csv("data/top20_players_raw.csv")
```

Data quality inspection

Before proceeding to the main analysis, we inspect the cleaned data for any remaining issues:

- verify the number of missing values
- check for duplicate matches
- identify any rows with incomplete information

This ensures the dataset is reliable for the research questions.

In [53]:
```
# Basic data checks
print("\nMatches columns:", matches.columns.tolist())
print("Stats columns:", stats.columns.tolist())
print("Players columns:", players.columns.tolist())

print("\nMatches dtypes:")
```

```
print(matches.dtypes.to_string())

print("\nMissing values in matches:")
print(matches.isna().sum())
```

```
Matches columns: ['date', 'tournament', 'surface', 'Round', 'winner', 'winner_r
ank_at_match', 'loser', 'loser_rank_at_match', 'score', 'source_player']
Stats columns: ['player', 'surface', 'match_record', 'match_wins', 'match_losse
s', 'match_pct', 'tiebreak_record', 'tiebreak_wins', 'tiebreak_losses', 'tiebre
ak_pct', 'ace_rate_pct', 'first_serve_in_pct', 'first_serve_points_won_pct', 's
econd_serve_points_won_pct', 'service_games_held_pct', 'service_points_won_pc
t', 'return_games_won_pct', 'return_points_won_pct', 'total_points_won_pct', 'd
ominance_ratio']
Players columns: ['name', 'age', 'current_rank']

Matches dtypes:
date                        str
tournament                  str
surface                     str
Round                       str
winner                      str
winner_rank_at_match    float64
loser                       str
loser_rank_at_match     float64
score                       str
source_player               str

Missing values in matches:
date                    0
tournament              0
surface                 0
Round                   0
winner                  0
winner_rank_at_match    0
loser                   0
loser_rank_at_match     1
score                   0
source_player           0
dtype: int64
```

In [54]:
```
missing = matches[matches["winner_rank_at_match"].isna() | matches["loser_rank

print("Total missing rows:", len(missing))
print(missing[["date", "source_player", "winner", "loser", "score", "winner_ra

print("\nCounts by source_player:")
print(missing["source_player"].value_counts())
```

```
Total missing rows: 1
           date     source_player            winner     loser    score  \
712  6-Feb-2026  Alexander Bublik  Alexander Bublik  Hugo Nys  6-0 6-3

     winner_rank_at_match  loser_rank_at_match
712                  10.0                  NaN

Counts by source_player:
source_player
Alexander Bublik    1
Name: count, dtype: int64
```

In [55]:
```python
print("\nStats dtypes:")
print(stats.dtypes.to_string())
```

```
Stats dtypes:
player                            str
surface                           str
match_record                      str
match_wins                      int64
match_losses                    int64
match_pct                         str
tiebreak_record                   str
tiebreak_wins                   int64
tiebreak_losses                 int64
tiebreak_pct                      str
ace_rate_pct                      str
first_serve_in_pct                str
first_serve_points_won_pct        str
second_serve_points_won_pct       str
service_games_held_pct            str
service_points_won_pct            str
return_games_won_pct              str
return_points_won_pct             str
total_points_won_pct              str
dominance_ratio                 float64
```

In [56]:
```python
print("\nMissing values in stats:")
print(stats.isna().sum())
```

```
Missing values in stats:
player                          0
surface                         0
match_record                    0
match_wins                      0
match_losses                    0
match_pct                       1
tiebreak_record                 0
tiebreak_wins                   0
tiebreak_losses                 0
tiebreak_pct                    5
ace_rate_pct                    1
first_serve_in_pct              1
first_serve_points_won_pct      1
second_serve_points_won_pct     1
service_games_held_pct          1
service_points_won_pct          1
return_games_won_pct            1
return_points_won_pct           1
total_points_won_pct            1
dominance_ratio                 1
dtype: int64
```

In [57]:

```
with pd.option_context('display.max_rows', None, 'display.max_columns', None,
    print(stats[stats.isna().any(axis=1)].to_string())
```

```
             player surface match_record  match_wins  match_losses match_pct ti
ebreak_record  tiebreak_wins  tiebreak_losses tiebreak_pct ace_rate_pct first_s
erve_in_pct first_serve_points_won_pct second_serve_points_won_pct service_game
s_held_pct service_points_won_pct return_games_won_pct return_points_won_pct to
tal_points_won_pct  dominance_ratio
19  Lorenzo Musetti   Grass     0-1 (0%)           0             1        0%
0-0 (-)              0                0          NaN         5.6%
58.9%                      61.9%                       61.4%                  7
2.2%                  61.7%                 5.3%                 29.7%
45.4%             0.78
38  Daniil Medvedev    Clay    4-4 (50%)           4             4       50%
0-0 (-)              0                0          NaN         6.6%
60.1%                      71.2%                       48.7%                  7
6.9%                  62.2%                25.3%                 38.7%
50.3%             1.02
47      Casper Ruud   Grass      0-0 (-)           0             0       NaN
0-0 (-)              0                0          NaN          NaN
NaN                        NaN                         NaN                    N
aN                    NaN                  NaN                    NaN
NaN              NaN
71       Tommy Paul   Grass    1-2 (33%)           1             2       33%
0-0 (-)              0                0          NaN         7.1%
56.4%                      76.1%                       53.7%                  8
3.3%                  66.3%                22.4%                 37.0%
51.2%             1.10
79   Frances Tiafoe   Grass    1-2 (33%)           1             2       33%
0-0 (-)              0                0          NaN        12.5%
51.6%                      74.5%                       47.8%                  7
7.3%                  61.6%                20.5%                 34.5%
48.4%             0.90
```

In [58]:
```
print("\nDuplicate match rows:", matches.duplicated(
   subset=["date", "tournament", "surface", "Round", "winner", "loser", "score
).sum())
```

```
Duplicate match rows: 133
```

In [59]:
```
# Inspect whether duplicates are exact copies, just from different sources (2
duplicates = matches[matches.duplicated(subset=["date", "tournament", "surface

exact_duplicate_counts = duplicates.groupby(
    ["date", "tournament", "surface", "Round", "winner", "loser", "score"]
).size().sort_values(ascending=False).tail(30)

print("\nTop duplicate groups by count:")
print(exact_duplicate_counts)
```

Top duplicate groups by count:

date	tournament	surface	Round	winner	loser	score	
4-Mar-2026	Indian Wells Masters	Hard	SF	Jannik Sinner	Alexander Zverev	6-2 6-4	2
5-Apr-2026	Monte-Carlo	Clay	R16	Alexander Bublik	Jiri Lehecka	6-2 7-5	2
5-Jan-2026	Hong Kong	Hard	F	Alexander Bublik	Lorenzo Musetti	7-6(2) 6-3	2
			SF	Lorenzo Musetti	Andrey Rublev	6-7(3) 7-5 6-4	2
7-Aug-2025	Cincinnati Masters	Hard	F	Carlos Alcaraz	Jannik Sinner	5-0 RET	2
			QF	Alexander Zverev	Ben Shelton	6-2 6-2	2
				Carlos Alcaraz	Andrey Rublev	6-3 4-6 7-5	2
			R16	Alexander Zverev	Karen Khachanov	7-5 3-0 RET	2
				Ben Shelton	Jiri Lehecka	6-4 6-4	2
			SF	Carlos Alcaraz	Alexander Zverev	6-4 6-3	2
7-May-2025	Rome Masters	Clay	F	Carlos Alcaraz	Jannik Sinner	7-6(5) 6-1	2
			QF	Jannik Sinner	Casper Ruud	6-0 6-1	2
				Lorenzo Musetti	Alexander Zverev	7-6(1) 6-4	2
			R16	Carlos Alcaraz	Karen Khachanov	6-3 3-6 7-5	2
				Jannik Sinner	Francisco Cerundolo	7-6(2) 6-3	2
				Lorenzo Musetti	Daniil Medvedev	7-5 6-4	2
			R64	Casper Ruud	Alexander Bublik	6-4 4-6 6-3	2
			SF	Carlos Alcaraz	Lorenzo Musetti	6-3 7-6(4)	2
				Jannik Sinner	Tommy Paul	1-6 6-0 6-3	2
9-Feb-2026	Dallas	Hard	F	Ben Shelton	Taylor Fritz	3-6 6-3 7-5	2
9-Jun-2025	Stuttgart	Grass	F	Taylor Fritz	Alexander Zverev	6-3 7-6(0)	2
			QF	Ben Shelton	Jiri Lehecka	6-4 6-4	2
			SF	Alexander Zverev	Ben Shelton	7-6(8) 7-6(1)	2
9-Nov-2025	Tour Finals	Hard	F	Jannik Sinner	Carlos Alcaraz	7-6(4) 7-5	2
			RR	Alexander Zverev	Ben Shelton	6-3 7-6(6)	2
				Carlos Alcaraz	Lorenzo Mus		

```
etti      6-4 6-1           2
                                                                    Taylor Frit
z         6-7(2) 7-5 6-3    2
                                                  Jannik Sinner     Alexander Z
verev     6-4 6-3           2
                                                                    Ben Shelton
6-3 7-6(3)        2
                                                  Taylor Fritz      Lorenzo Mus
etti      6-3 6-4           2
dtype: int64
```

Note

1. Missing rank values in matches: loser was unranked at match time
2. Missing stat values (NaN) in stats: player never played on that surface (Casper Ruud on grass)
3. Missing tiebreak (NaN) in stats: the player played on the surface, but no tiebreaks were played. These are intentionally left as NaN to avoid implying zero performance.
4. Duplicate matches: same match scraped twice, once per player perspective
5. Percentage columns read as strings due to % suffix (e.g. "79.8%")

Actions taken:

- Duplicates removed keeping first occurrence
- Parse percentage columns from string (e.g. "79.8%") to float
- Preserved NaN values for:
    - players with no surface matches
    - players with no observed tiebreaks

  to maintain semantic correctness and avoid analytical bias

In [60]:
```
matches = matches.drop_duplicates(subset=["date", "tournament", "surface", "Ro
# Use the deduplicated version for analysis
```

In [61]:
```
# Parse stats percentages
percentage_columns = [
    "match_pct", "tiebreak_pct", "ace_rate_pct", "first_serve_in_pct", "first_
    "second_serve_points_won_pct", "service_games_held_pct", "service_points_w
    "return_games_won_pct", "return_points_won_pct", "total_points_won_pct",
    "dominance_ratio"
]

for col in percentage_columns:
    if col in stats.columns:
```

```
        stats[col] = pd.to_numeric(
            stats[col].astype(str).str.rstrip("%"),
            errors="coerce",
        ).astype("float64")

stats.head(5)
```

Out[61]:

	player	surface	match_record	match_wins	match_losses	match_pct	tiebrea
0	Carlos Alcaraz	Overall last 52	70-7 (91%)	70	7	91.0	20
1	Carlos Alcaraz	Hard	40-5 (89%)	40	5	89.0	1
2	Carlos Alcaraz	Clay	19-1 (95%)	19	1	95.0	
3	Carlos Alcaraz	Grass	11-1 (92%)	11	1	92.0	
4	Jannik Sinner	Overall last 52	72-8 (90%)	72	8	90.0	1

Data validation

We check:

- verify no duplicate match rows remain
- confirm stats percentage fields are floats

In [62]:

```
print("\nDuplicate rows after validation:", matches.duplicated(
    subset=["date", "tournament", "surface", "Round", "winner", "loser", "scor
).sum())
print("\nStats dtypes:")
print(stats.dtypes.to_string())
```

```
Duplicate rows after validation: 0

Stats dtypes:
player                            str
surface                           str
match_record                      str
match_wins                      int64
match_losses                    int64
match_pct                     float64
tiebreak_record                   str
tiebreak_wins                   int64
tiebreak_losses                 int64
tiebreak_pct                  float64
ace_rate_pct                  float64
first_serve_in_pct            float64
first_serve_points_won_pct    float64
second_serve_points_won_pct   float64
service_games_held_pct        float64
service_points_won_pct        float64
return_games_won_pct          float64
return_points_won_pct         float64
total_points_won_pct          float64
dominance_ratio               float64
```

Q1 — How important is ranking for match outcomes?

We analyse whether the higher-ranked player (lower rank number) tends to win, and whether the size of the ranking gap affects the probability of an upset.

Steps:

1. Compute the rank difference (loser_rank − winner_rank) for each match
2. Label each match as "favourite won" or "upset"
3. Bin matches by rank gap and compute upset rate per bin
4. Visualise results

Note: Matches involving players without an official ranking at match time were excluded from this analysis, as rank differences and favourite status cannot be defined for these cases.

In [63]:
```
# Drop the one match with missing loser rank
matches_ranked = matches.dropna(subset=["winner_rank_at_match", "loser_rank_at

# Rank difference: positive means the winner was ranked higher (better) than t
matches_ranked["rank_diff"] = matches_ranked["loser_rank_at_match"] - matches_

# Favourite = the player with the lower rank number (better ranked) going into
matches_ranked["favourite_won"] = matches_ranked["rank_diff"] > 0
```

```
overall_fav_rate = matches_ranked["favourite_won"].mean() * 100
print(f"Matches analysed: {len(matches_ranked)}")
print(f"Favourite won: {matches_ranked['favourite_won'].sum()} ({overall_fav_r
print(f"Upsets: {(~matches_ranked['favourite_won']).sum()} ({100 - overall_fav
```

```
Matches analysed: 1165
Favourite won: 860 (73.8%)
Upsets: 305 (26.2%)
```

In [64]:

```
# Drop the one match with missing loser rank
matches_ranked = matches.dropna(subset=["winner_rank_at_match", "loser_rank_at

# Rank difference: positive means the winner was ranked higher (better) than t
matches_ranked["rank_diff"] = matches_ranked["loser_rank_at_match"] - matches_

# Favourite = the player with the lower rank number (better ranked) going into
matches_ranked["favourite_won"] = matches_ranked["rank_diff"] > 0

overall_fav_rate = matches_ranked["favourite_won"].mean() * 100
print(f"Matches analysed: {len(matches_ranked)}")
print(f"Favourite won: {matches_ranked['favourite_won'].sum()} ({overall_fav_r
print(f"Upsets: {(~matches_ranked['favourite_won']).sum()} ({100 - overall_fav
```

```
Matches analysed: 1165
Favourite won: 860 (73.8%)
Upsets: 305 (26.2%)
```

In [65]:

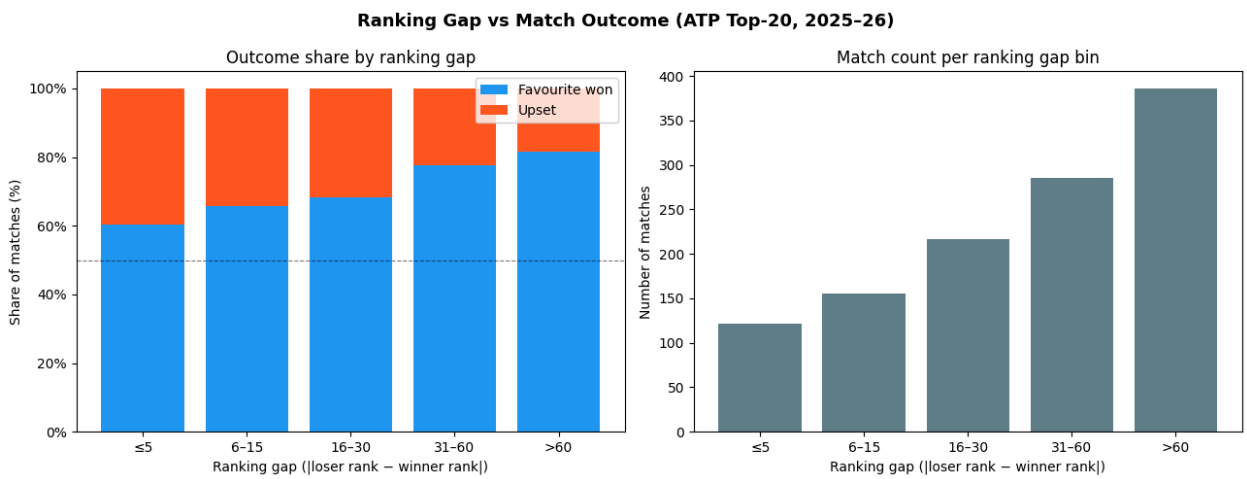
```
# Use absolute rank gap for binning regardless of who won
matches_ranked["rank_gap"] = matches_ranked["rank_diff"].abs()

bins   = [0, 5, 15, 30, 60, matches_ranked["rank_gap"].max() + 1]
labels = ["≤5", "6–15", "16–30", "31–60", ">60"]
matches_ranked["gap_bin"] = pd.cut(matches_ranked["rank_gap"], bins=bins, labe

bin_stats = (
    matches_ranked
    .groupby("gap_bin", observed=True)["favourite_won"]
    .agg(total="count", fav_wins="sum")
    .assign(fav_win_pct=lambda d: d["fav_wins"] / d["total"] * 100,
            upset_pct=lambda d: 100 - d["fav_wins"] / d["total"] * 100)
    .reset_index()
)

print(bin_stats.to_string(index=False))
```

```
gap_bin  total  fav_wins  fav_win_pct  upset_pct
     ≤5    121        73    60.330579  39.669421
   6–15    155       102    65.806452  34.193548
  16–30    217       148    68.202765  31.797235
  31–60    286       222    77.622378  22.377622
    >60    386       315    81.606218  18.393782
```

In [66]:

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
fig.suptitle("Ranking Gap vs Match Outcome (ATP Top-20, 2025–26)", fontsize=13
```

```
# --- left: stacked bar (favourite vs upset share) ---
ax = axes[0]
ax.bar(bin_stats["gap_bin"], bin_stats["fav_win_pct"], label="Favourite won",
ax.bar(bin_stats["gap_bin"], bin_stats["upset_pct"],   label="Upset",
       bottom=bin_stats["fav_win_pct"])
ax.axhline(50, color="black", linewidth=0.8, linestyle="--", alpha=0.5)
ax.set_xlabel("Ranking gap (|loser rank − winner rank|)")
ax.set_ylabel("Share of matches (%)")
ax.set_title("Outcome share by ranking gap")
ax.yaxis.set_major_formatter(mtick.PercentFormatter())
ax.legend()

# --- right: match count per bin ---
ax2 = axes[1]
ax2.bar(bin_stats["gap_bin"], bin_stats["total"], color="#607D8B")
ax2.set_xlabel("Ranking gap (|loser rank − winner rank|)")
ax2.set_ylabel("Number of matches")
ax2.set_title("Match count per ranking gap bin")

plt.tight_layout()
plt.savefig(os.path.join(PLOTS_DIR,"q1_ranking_vs_outcome.png"), dpi=150, bbox
plt.show()
```



Findings

- Overall, the higher-ranked player won **73.8% overall** of matches, confirming that ranking is a meaningful predictor of match outcome.
- The upset rate is highest in the **≤5** gap bin, where both players are nearly equally ranked, and drops as the gap widens.
- Matches with a gap greater than 60 ranks are almost always won by the favourite (~81.6% favourite wins), suggesting that large ranking differences are very reliable predictors.
- However, even with large gaps upsets do occur, which reflects the high-variance nature of individual tennis matches.

Q2 — Do players perform differently depending on the surface?

We analyse whether top players show surface specialisation or perform consistently across Hard, Clay and Grass.

Steps:

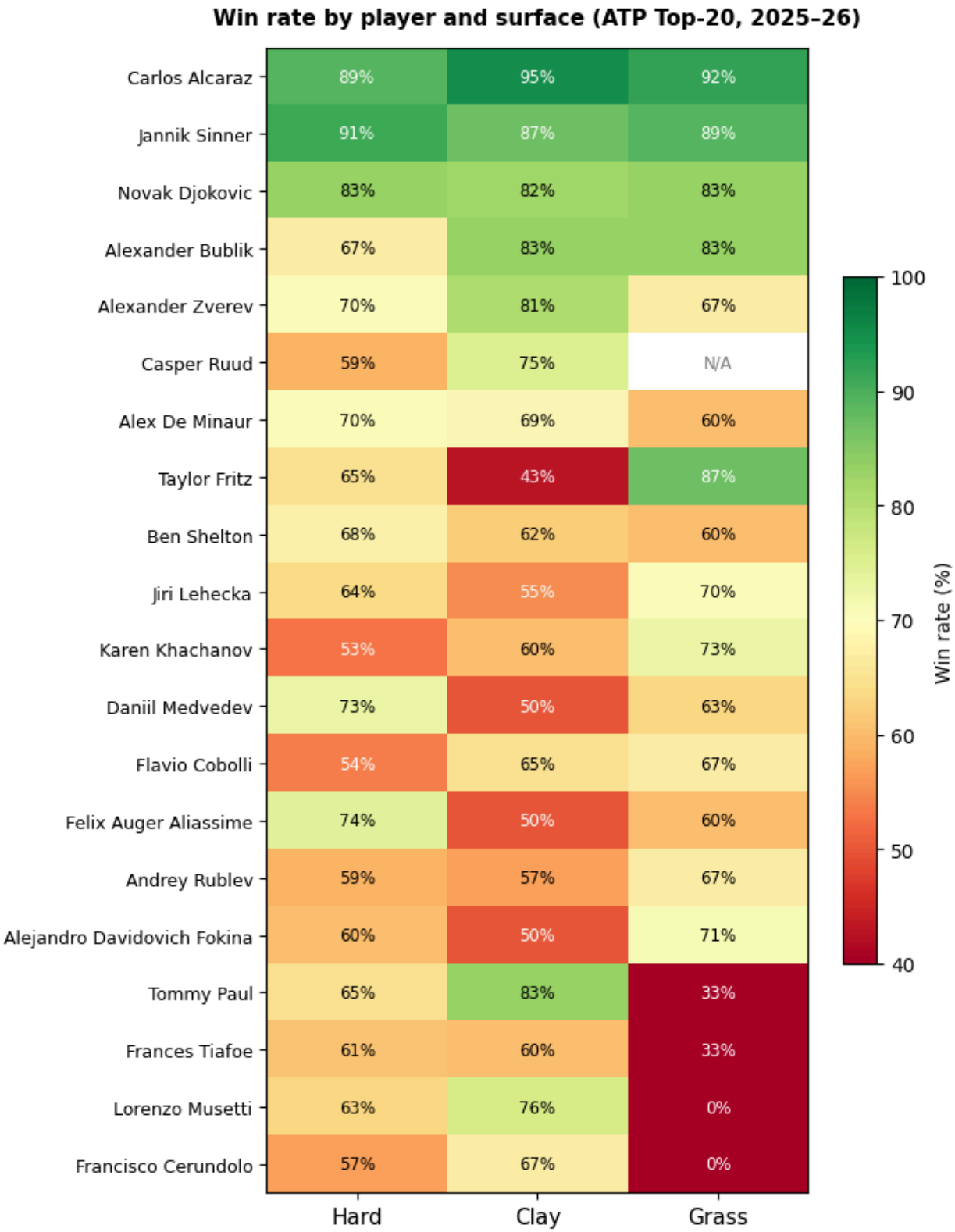
1. Filter stats to the three main surfaces (exclude "Overall last 52")
2. Compare win rate (match_pct) and dominance ratio per player per surface
3. Identify the most and least consistent players across surfaces
4. Visualise with a heatmap and grouped bar chart

In [67]:
```
# Keep only the three main surfaces
surfaces = ["Hard", "Clay", "Grass"]
stats_surfaces = stats[stats["surface"].isin(surfaces)].copy()

# Ensure match_pct is numeric (coerce invalid to NaN)
stats_surfaces["match_pct"] = (
    stats_surfaces["match_pct"]
    .astype(str)
    .str.replace("%", "", regex=False)
    .replace("-", np.nan)
    .astype(float)
)

# Pivot: one row per player, one column per surface for each metric
pct_pivot = stats_surfaces.pivot(index="player", columns="surface", values="ma
dr_pivot  = stats_surfaces.pivot(index="player", columns="surface", values="dc

# Surface consistency: std across the three surfaces (lower = more consistent)
pct_pivot["std"] = pct_pivot[surfaces].std(axis=1)
pct_pivot["surfaces_played"] = pct_pivot[surfaces].notna().sum(axis=1)
pct_pivot_strict = pct_pivot[pct_pivot["surfaces_played"] == 3]

print("Win rate (match_pct) by player and surface:")
print(pct_pivot.sort_values("std", ascending=False).to_string())
```

```
Win rate (match_pct) by player and surface:
surface                      Hard  Clay  Grass        std  surfaces_played
player
Lorenzo Musetti              63.0  76.0    0.0  40.648903                3
Francisco Cerundolo          57.0  67.0    0.0  36.143234                3
Tommy Paul                   65.0  83.0   33.0  25.324560                3
Taylor Fritz                 65.0  43.0   87.0  22.000000                3
Frances Tiafoe               61.0  60.0   33.0  15.885003                3
Felix Auger Aliassime        74.0  50.0   60.0  12.055428                3
Daniil Medvedev              73.0  50.0   63.0  11.532563                3
Casper Ruud                  59.0  75.0    NaN  11.313708                2
Alejandro Davidovich Fokina  60.0  50.0   71.0  10.503968                3
Karen Khachanov              53.0  60.0   73.0  10.148892                3
Alexander Bublik             67.0  83.0   83.0   9.237604                3
Jiri Lehecka                 64.0  55.0   70.0   7.549834                3
Alexander Zverev             70.0  81.0   67.0   7.371115                3
Flavio Cobolli               54.0  65.0   67.0   7.000000                3
Alex De Minaur               70.0  69.0   60.0   5.507571                3
Andrey Rublev                59.0  57.0   67.0   5.291503                3
Ben Shelton                  68.0  62.0   60.0   4.163332                3
Carlos Alcaraz               89.0  95.0   92.0   3.000000                3
Jannik Sinner                91.0  87.0   89.0   2.000000                3
Novak Djokovic               83.0  82.0   83.0   0.577350                3
```

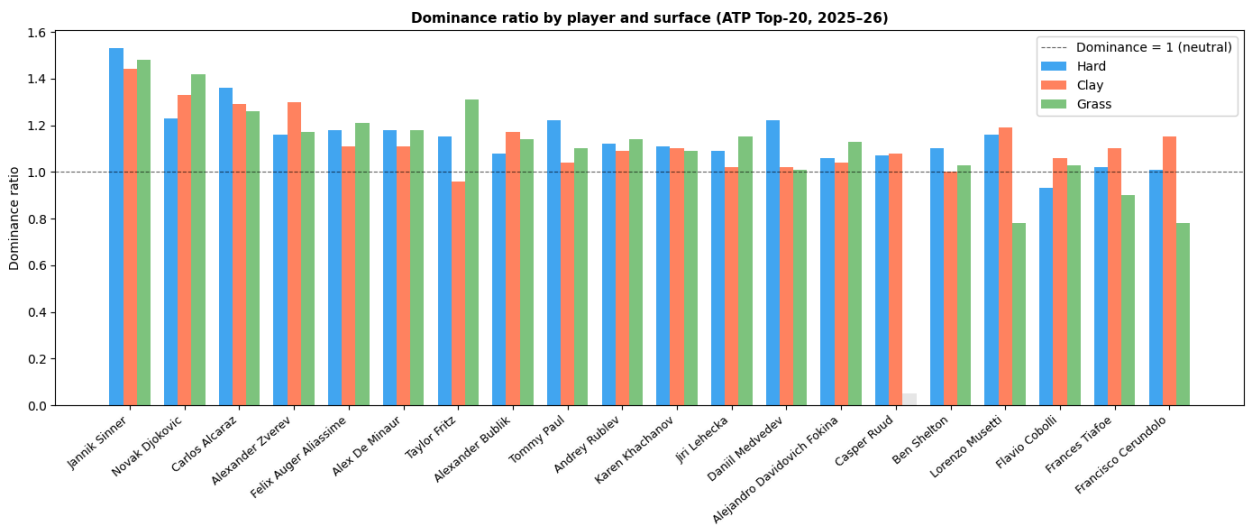
In [68]:

```
# Sort players by overall Hard court win rate for a readable order
plot_order = pct_pivot[surfaces].mean(axis=1).sort_values(ascending=False).inc
heatmap_data = pct_pivot.loc[plot_order, surfaces]

fig, ax = plt.subplots(figsize=(7, 9))

im = ax.imshow(heatmap_data.values, aspect="auto", cmap="RdYlGn", vmin=40, vma

# Axis labels
ax.set_xticks(range(len(surfaces)))
ax.set_xticklabels(surfaces, fontsize=11)
ax.set_yticks(range(len(heatmap_data)))
ax.set_yticklabels(heatmap_data.index, fontsize=9)

# Annotate each cell with the value or "N/A"
for i, player in enumerate(heatmap_data.index):
    for j, surface in enumerate(surfaces):
        val = heatmap_data.loc[player, surface]
        text = f"{val:.0f}%" if not np.isnan(val) else "N/A"
        color = "black" if 55 < val < 85 else "white" if not np.isnan(val) els
        ax.text(j, i, text, ha="center", va="center", fontsize=8.5, color=colc

plt.colorbar(im, ax=ax, label="Win rate (%)", shrink=0.6)
ax.set_title("Win rate by player and surface (ATP Top-20, 2025–26)",
             fontsize=11, fontweight="bold", pad=12)
plt.tight_layout()
plt.savefig(os.path.join(PLOTS_DIR,"q2_winrate_heatmap.png"), dpi=150, bbox_ir
plt.show()
```



In [69]:
```
players_ordered = dr_pivot[surfaces].mean(axis=1).sort_values(ascending=False)
dr_plot = dr_pivot.loc[players_ordered, surfaces]

x = np.arange(len(players_ordered))
width = 0.25
```

```
colors = {"Hard": "#2196F3", "Clay": "#FF7043", "Grass": "#66BB6A"}

fig, ax = plt.subplots(figsize=(14, 6))

for i, surface in enumerate(surfaces):
    vals = dr_plot[surface].values
    bars = ax.bar(x + i * width, vals, width, label=surface,
                  color=colors[surface], alpha=0.85)

    # Grey out NaN bars visually
    for j, v in enumerate(vals):
        if np.isnan(v):
            ax.bar(x[j] + i * width, 0.05, width, color="lightgrey", alpha=0.5

ax.axhline(1.0, color="black", linewidth=0.8, linestyle="--", alpha=0.6, label
ax.set_xticks(x + width)
ax.set_xticklabels(players_ordered, rotation=40, ha="right", fontsize=9)
ax.set_ylabel("Dominance ratio")
ax.set_title("Dominance ratio by player and surface (ATP Top-20, 2025–26)",
             fontsize=11, fontweight="bold")
ax.legend()
plt.tight_layout()
plt.savefig(os.path.join(PLOTS_DIR,"q2_dominance_ratio.png"), dpi=150, bbox_in
plt.show()
```



Note

For consistency comparisons, we restrict the analysis to players with match data on all three surfaces, since zero values for grass reflect lack of participation rather than poor performance.

In [70]:
```
# Standard deviation of match_pct across surfaces (ignoring NaN)
consistency = (
    stats_surfaces
    .groupby("player")["match_pct"]
```

```
    .agg(
        win_rate_std="std",
        surfaces_played="count"
    )
    .query("surfaces_played == 3")
    .sort_values("win_rate_std")
    .reset_index()
)

print("Surface consistency ranking (lower std = more consistent across surface
print(consistency.to_string(index=False))
```

```
Surface consistency ranking (lower std = more consistent across surfaces):
                     player  win_rate_std  surfaces_played
             Novak Djokovic      0.577350                3
              Jannik Sinner      2.000000                3
             Carlos Alcaraz      3.000000                3
                Ben Shelton      4.163332                3
               Andrey Rublev      5.291503                3
             Alex De Minaur      5.507571                3
             Flavio Cobolli      7.000000                3
           Alexander Zverev      7.371115                3
               Jiri Lehecka      7.549834                3
           Alexander Bublik      9.237604                3
            Karen Khachanov     10.148892                3
Alejandro Davidovich Fokina     10.503968                3
            Daniil Medvedev     11.532563                3
      Felix Auger Aliassime     12.055428                3
             Frances Tiafoe     15.885003                3
               Taylor Fritz     22.000000                3
                 Tommy Paul     25.324560                3
        Francisco Cerundolo     36.143234                3
            Lorenzo Musetti     40.648903                3
```

Findings

- The top tier is clearly split into **all-surface elites** and **surface-tilted performers**. **Carlos Alcaraz, Jannik Sinner, and Novak Djokovic** remain high on all three surfaces (roughly 82-95% win rate), indicating broad adaptability rather than one-surface dependence.
- The strongest clay specialists in this sample are visible in both metrics: players like **Alcaraz** and **Djokovic** combine very high clay win rates with clay dominance ratios comfortably above 1.0. This suggests their clay success is backed by point-level control, not only close-match variance.
- Grass shows the largest dispersion across players. Some players post very high grass outcomes (for example **Alcaraz** and **Sinner**), while others are at the bottom end of the distribution.
- **Taylor Fritz** and **Tommy Paul** are clear examples of strong surface

sensitivity. Fritz drops sharply on clay but is much stronger on grass, while Paul is very strong on clay but drops heavily on grass. Because this pattern appears in both **win rate** and **dominance ratio**, it looks like a genuine surface-fit effect rather than random short-term fluctuation.

- In this project interpretation, **Lorenzo Musetti** and **Francisco Cerundolo** are treated as not having meaningful grass participation in the scraped period; therefore they appear at **0% on grass** and this should not be interpreted as a direct quality ranking against players with full grass samples.
- The **dominance ratio** chart broadly confirms the win-rate heatmap: where a player has higher win rate on a surface, their dominance ratio is usually also higher there. This cross-metric alignment strengthens the interpretation that differences are structural (serve/return balance), not random noise.
- There are important exceptions worth noting. A player can be competitive in win rate while showing only moderate dominance ratio (or vice versa), which typically points to match-context effects (sample size, close-match distribution, opponent mix) rather than a pure quality gap.
- **Casper Ruud** has missing grass values (no grass matches in the scraped period), so his cross-surface consistency cannot be interpreted symmetrically with players who have data on all three surfaces. Comparisons of consistency should therefore prioritise complete three-surface profiles.

Q3 — Which player characteristics are associated with winning matches?

We join match results with per-surface stats and player profile data (age, rank) to compare the characteristics of winners vs losers at the point of each match.

Steps:

1. Join matches with stats on (player, surface) for both winner and loser
2. Join player age from the players table
3. Compare key stats between winners and losers
4. Visualise distributions with box plots and a correlation heatmap

Note on missing values: Winner–loser comparisons are computed only for matches where both players have surface-specific statistics available. As a result, metrics

such as tiebreak win percentage are based on smaller subsets of matches and should be interpreted as conditional effects, rather than global predictors across all matches.

In [71]:
```
# We only join surface-specific stats (not "Overall last 52")
surfaces = ["Hard", "Clay", "Grass"]
stats_for_join = stats[stats["surface"].isin(surfaces)].copy()

# Columns we want to carry into the match rows
stat_cols = [
    "ace_rate_pct",
    "first_serve_in_pct",
    "first_serve_points_won_pct",
    "second_serve_points_won_pct",
    "service_games_held_pct",
    "return_games_won_pct",
    "return_points_won_pct",
    "dominance_ratio",
    "tiebreak_pct",
]

stats_slim = stats_for_join[["player", "surface"] + stat_cols].copy()
```

In [72]:
```
# --- winner stats ---
winner_stats = (
    stats_slim
    .rename(columns={c: f"w_{c}" for c in stat_cols})
    .rename(columns={"player": "winner", "surface": "surface"})
)

# --- loser stats ---
loser_stats = (
    stats_slim
    .rename(columns={c: f"l_{c}" for c in stat_cols})
    .rename(columns={"player": "loser", "surface": "surface"})
)

# --- age from players table ---
# Normalise name format just in case of whitespace issues
players["name"] = players["name"].str.strip()

winner_age = players[["name", "age"]].rename(columns={"name": "winner", "age":
loser_age  = players[["name", "age"]].rename(columns={"name": "loser",  "age":

# --- build enriched matches ---
enriched = (
    matches
    .merge(winner_stats, on=["winner", "surface"], how="left")
    .merge(loser_stats,  on=["loser",  "surface"], how="left")
    .merge(winner_age,   on="winner",              how="left")
    .merge(loser_age,    on="loser",               how="left")
)
```

```
print(f"Enriched matches shape: {enriched.shape}")
print(f"Winner stats coverage : {enriched['w_dominance_ratio'].notna().sum()}
print(f"Loser stats coverage  : {enriched['l_dominance_ratio'].notna().sum()}
print(f"Winner age coverage   : {enriched['winner_age'].notna().sum()} / {len(
```

```
Enriched matches shape: (1166, 30)
Winner stats coverage : 823 / 1166
Loser stats coverage  : 391 / 1166
Winner age coverage   : 823 / 1166
```

In [73]:

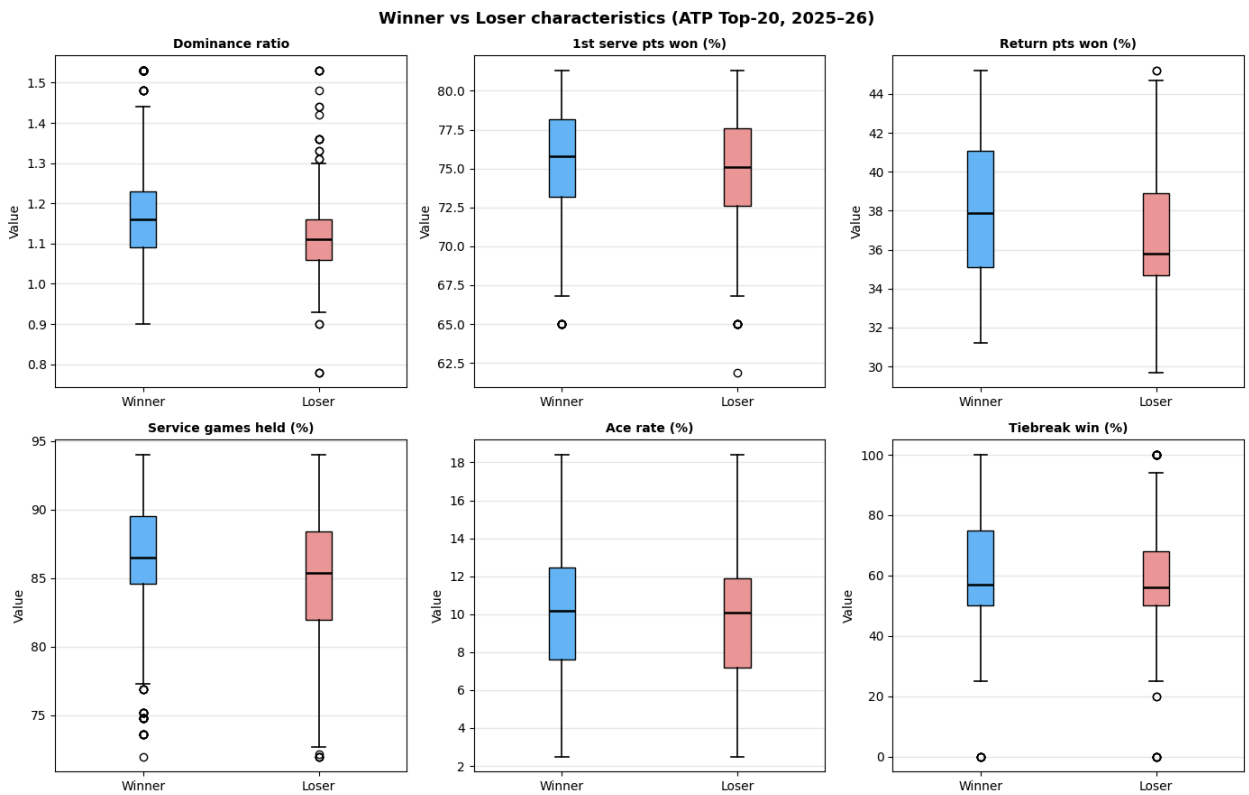
```
# Delta = winner stat - loser stat for each metric
# A positive delta means the winner was better on that characteristic
delta_cols = {}
for col in stat_cols:
    delta_name = f"delta_{col}"
    enriched[delta_name] = enriched[f"w_{col}"] - enriched[f"l_{col}"]
    delta_cols[col] = delta_name

enriched["delta_rank"] = enriched["loser_rank_at_match"] - enriched["winner_ra
enriched["delta_age"]  = enriched["winner_age"] - enriched["loser_age"]

print("Delta columns added:")
for orig, delta in delta_cols.items():
    mean_val = enriched[delta].mean()
    print(f"  {delta:45s}  mean = {mean_val:+.3f}")
```

```
Delta columns added:
  delta_ace_rate_pct                            mean = +0.541
  delta_first_serve_in_pct                      mean = +1.460
  delta_first_serve_points_won_pct              mean = +1.140
  delta_second_serve_points_won_pct             mean = +1.400
  delta_service_games_held_pct                  mean = +2.646
  delta_return_games_won_pct                    mean = +3.117
  delta_return_points_won_pct                   mean = +1.584
  delta_dominance_ratio                         mean = +0.110
  delta_tiebreak_pct                            mean = +4.495
```

In [74]:

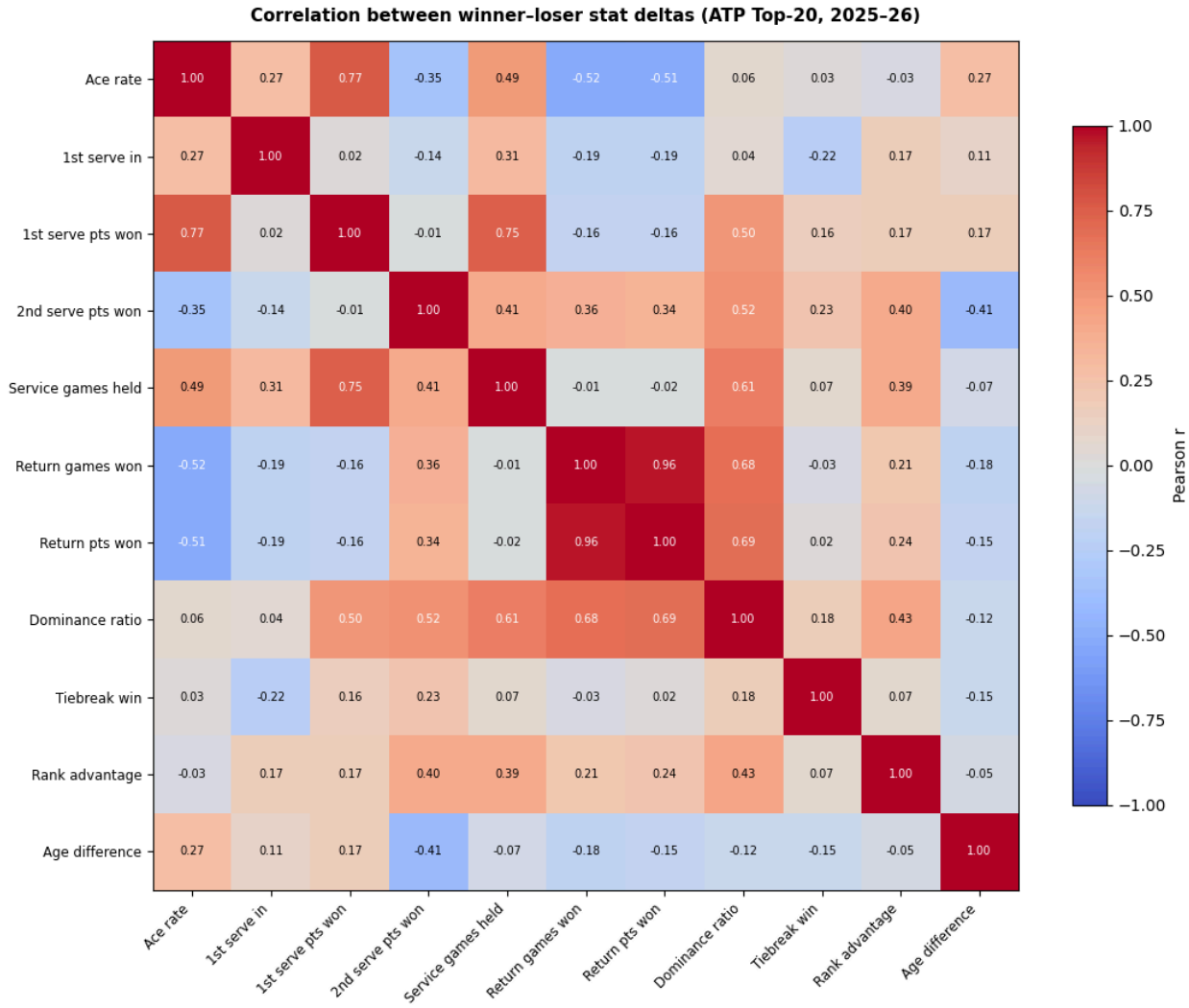
```
plot_stats = [
    ("w_dominance_ratio",            "l_dominance_ratio",            "Dominanc
    ("w_first_serve_points_won_pct", "l_first_serve_points_won_pct", "1st serv
    ("w_return_points_won_pct",      "l_return_points_won_pct",      "Return p
    ("w_service_games_held_pct",     "l_service_games_held_pct",     "Service
    ("w_ace_rate_pct",               "l_ace_rate_pct",               "Ace rate
    ("w_tiebreak_pct",               "l_tiebreak_pct",               "Tiebreak
]

fig, axes = plt.subplots(2, 3, figsize=(14, 9))
axes = axes.flatten()

for ax, (w_col, l_col, title) in zip(axes, plot_stats):
    w_vals = enriched[w_col].dropna()
    l_vals = enriched[l_col].dropna()
```

```
    bp = ax.boxplot(
        [w_vals, l_vals],
        tick_labels=["Winner", "Loser"],
        patch_artist=True,
        medianprops=dict(color="black", linewidth=1.8),
        whiskerprops=dict(linewidth=1.2),
        capprops=dict(linewidth=1.2),
    )
    bp["boxes"][0].set_facecolor("#64B5F6")   # winner = blue
    bp["boxes"][1].set_facecolor("#EF9A9A")   # loser  = red

    ax.set_title(title, fontsize=10, fontweight="bold")
    ax.set_ylabel("Value")
    ax.grid(axis="y", alpha=0.3)

fig.suptitle("Winner vs Loser characteristics (ATP Top-20, 2025–26)",
             fontsize=13, fontweight="bold")
plt.tight_layout()
plt.savefig(os.path.join(PLOTS_DIR,"q3_winner_vs_loser_boxplots.png"), dpi=150
plt.show()
```



In [75]:
```
# Build a dataframe of all deltas and correlate with a binary outcome (1 = fav
all_delta_cols = list(delta_cols.values()) + ["delta_rank", "delta_age"]

delta_df = enriched[all_delta_cols].copy()

# Readable labels for the plot
label_map = {
    "delta_ace_rate_pct":                    "Ace rate",
```

```
    "delta_first_serve_in_pct":            "1st serve in",
    "delta_first_serve_points_won_pct":    "1st serve pts won",
    "delta_second_serve_points_won_pct":   "2nd serve pts won",
    "delta_service_games_held_pct":        "Service games held",
    "delta_return_games_won_pct":          "Return games won",
    "delta_return_points_won_pct":         "Return pts won",
    "delta_dominance_ratio":               "Dominance ratio",
    "delta_tiebreak_pct":                  "Tiebreak win",
    "delta_rank":                          "Rank advantage",
    "delta_age":                           "Age difference",
}

corr = delta_df.rename(columns=label_map).corr()

fig, ax = plt.subplots(figsize=(11, 9))
cax = ax.imshow(corr.values, cmap="coolwarm", vmin=-1, vmax=1, aspect="auto")
plt.colorbar(cax, ax=ax, shrink=0.8, label="Pearson r")

ticks = list(label_map.values())
ax.set_xticks(range(len(ticks)))
ax.set_xticklabels(ticks, rotation=45, ha="right", fontsize=8.5)
ax.set_yticks(range(len(ticks)))
ax.set_yticklabels(ticks, fontsize=8.5)

# Annotate cells
for i in range(len(ticks)):
    for j in range(len(ticks)):
        val = corr.values[i, j]
        ax.text(j, i, f"{val:.2f}", ha="center", va="center",
                fontsize=7, color="white" if abs(val) > 0.5 else "black")

ax.set_title("Correlation between winner–loser stat deltas (ATP Top-20, 2025–2
             fontsize=11, fontweight="bold", pad=12)
plt.tight_layout()
plt.savefig(os.path.join(PLOTS_DIR,"q3_delta_correlation_heatmap.png"), dpi=15
plt.show()
```



In [76]:

```
summary = (
    pd.DataFrame({
        "characteristic": list(label_map.values()),
        "mean_delta":     [enriched[c].mean() for c in all_delta_cols],
        "matches_with_data": [enriched[c].notna().sum() for c in all_delta_col
    })
    .sort_values("mean_delta", ascending=False)
    .reset_index(drop=True)
)

print("Mean delta (winner − loser) per characteristic:")
print(summary.to_string(index=False))
```

```
Mean delta (winner - loser) per characteristic:
    characteristic  mean_delta  matches_with_data
    Rank advantage   33.957082               1165
      Tiebreak win    4.494681                188
  Return games won    3.117368                190
Service games held    2.646316                190
    Return pts won    1.583684                190
      1st serve in    1.460000                190
 2nd serve pts won    1.400000                190
 1st serve pts won    1.140000                190
          Ace rate    0.541053                190
   Dominance ratio    0.110053                190
    Age difference   -0.878947                190
```

Findings — Player Characteristics Associated with Winning

Overall performance differences (Winner vs. Loser)

- **Dominance ratio** shows a clear and consistent separation between winners and losers.
    - Winners have systematically higher values, indicating that overall point superiority is one of the **strongest indicators of match success**.
- **First serve points won (%)** is noticeably higher for winners.
    - This suggests that **effectiveness on first serve (not just getting it in)** is a key differentiator.
- **Return points won (%)** also shows a meaningful gap.
    - Winners are not only strong on serve but also **consistently outperform opponents on return**, highlighting the importance of all-court performance.
- **Service games held (%)** is slightly higher for winners but with overlap.
    - This indicates that while holding serve is important, it is **not sufficient alone** to explain winning.
- **Ace rate (%)** shows only minor differences.
    - High ace rates are **not a decisive factor** among top players, suggesting that consistency and point construction matter more than pure power.
- **Tiebreak win (%)** distributions are highly variable with significant

overlap.

        - This metric appears **noisy and less reliable**, likely due to smaller sample sizes and situational dependence.

Insights from winner–loser deltas

- Positive deltas (winner − loser) are observed across most key metrics, confirming that winners are generally **better in multiple dimensions simultaneously**, rather than excelling in just one.
- The strongest average advantages appear in:
        - **First serve points won**
        - **Return performance**
        - **Dominance ratio**

Correlation analysis of performance gaps

- **Return games won (%) and return points won (%) are almost perfectly correlated (r ≈ 0.96)**
  → These metrics capture essentially the same underlying concept of **return strength**.
- **Dominance ratio strongly correlates with return performance (r ≈ 0.68–0.69)**
  → Indicates that **return ability is a major driver of overall match dominance**.
- **First serve points won (%) strongly correlates with service games held (%) (r ≈ 0.75)**
  → Confirms that **winning points on first serve directly translates into holding serve**.
- **Ace rate is strongly correlated with first serve points won (r ≈ 0.77)**
  → Suggests that aces contribute to serve effectiveness, but:
        - Given weak separation in boxplots, this effect is **not sufficient alone to explain match outcomes**.
- **Ace rate is negatively correlated with return metrics (≈ -0.5)**

→ Players who rely more on serve dominance tend to be **weaker on return**, indicating a trade-off in play styles.

- **Second serve points won (%) correlates moderately with dominance ratio (r ≈ 0.52)**
  → Highlights the importance of **consistency under pressure**, especially when the first serve fails.

Role of contextual factors

- **Rank advantage shows moderate positive correlation with dominance ratio (r ≈ 0.43)**
  → Higher-ranked players tend to dominate more, but rank alone does not fully explain match outcomes.

- **Age difference has weak correlations with most performance metrics**
  → Suggests that within the ATP Top-20, **age is not a decisive factor** for match success.

Key takeaway

- Winning at the top level is driven by a **combination of serve efficiency and return strength**, rather than isolated metrics.
- In particular:
  - **Return performance + dominance ratio** are the strongest overall indicators
  - **First serve effectiveness** is more important than raw serving power (aces)
- Success is therefore best explained by **balanced, all-around performance**, rather than specialization.